



Commodity-price comovement and global economic activity

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ABSTRACT

Guided by a macroeconomic model with endogenous commodity prices, we apply a new factor-based identification strategy to decompose the historical sources of changes in commodity prices and global economic activity. The model yields a factor structure for commodity prices and identification conditions that provide an economic interpretation: one factor captures the combined contribution of shocks that affect commodity markets only through general-equilibrium forces. Applied to a cross-section of commodity prices since 1968, the theoretical restrictions are consistent with the data and yield structural interpretations of the common factors in commodity prices. Commodity-related shocks have contributed modestly to global economic fluctuations.

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1. Introduction

Between January 2003 and July 2008, the prices of most major commodities grew rapidly: wheat by 120%, copper by 363%, aluminum by 100% and nickel by 138%. Many observers concluded that the simultaneous rise in prices across such a broad cross-section of commodities reflected a common cause—an increase in the global demand for commodities due to growth in emerging Asia and especially China. Other episodes of widespread comovement in commodity prices have similarly suggested that global demand is a common source of movements in commodity prices, such as the early 1970s or the late 1990s. But this explanation is not necessarily the only one: exogenous changes in the prices of oil and other energy products could simultaneously drive the prices of many non-energy commodities because of the important role played by transportation costs in their distribution. In addition, changing preferences on the part of consumers could shift the demand for commodity-intensive products, as could technological changes that affect the relative importance of raw materials in the production of consumption goods. Decomposing the sources of commodity price comovement is therefore inextricably linked to identifying the sources of global business cycle fluctuations.

In this paper, we develop and implement a new methodology for decomposing the sources of commodity price comovement and global business cycle fluctuations. Underlying this methodology is a general-equilibrium model of global business cycles with commodities that predicts a factor structure for real commodity prices. The predicted factor structure decomposes the sources of global business cycle fluctuations and commodity price movements, and the theory suggests several ways to recover a structural interpretation to the common factors extracted from commodity prices. In other words, this

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methodology provides a way to use the comovement in commodity prices to disentangle the simultaneous determination of commodity prices and business cycles.

The factor structure in commodity prices predicted by the model separates exogenous forces (or “shocks”) into two types. The first set of shocks includes those that directly shift the supply and demand curves for commodities and thus would affect commodity prices even in the absence of general-equilibrium changes in aggregate income, though such forces may also have general-equilibrium effects on aggregate income and therefore additional indirect effects on commodity prices. We refer to these factors as direct factors. They potentially reflect a variety of common shocks to the prices of inputs used to produce commodities, such as labor or energy, common productivity shocks, or demand factors such as changes in the relative need for commodities to produce final consumption goods. The second set of shocks includes those that affect commodity prices only indirectly through their effects on aggregate output. We refer to these as indirect factors. The indirect effects can come through two channels. One is the standard demand channel. When aggregate economic activity is high, the demand for commodities used to produce the final good is also high, thereby raising the prices of all commodities. The second is a supply-side channel. When aggregate income is high, agents may be less willing to supply the inputs used to produce commodities because of income effects, thereby pushing up the prices of commodities. Both channels induce positive comovement in the prices of commodities.

The theory predicts a new result about indirect shocks. Because their effects on commodity prices are summarized entirely by their effects on aggregate output, *each of the indirect shocks induces the same comovement among commodity prices*. As a result, their combined effect on commodity prices can be aggregated into a single factor. Furthermore, this factor has a precise structural interpretation in the model. It corresponds to the *counterfactual* level of global economic activity that would have been obtained without direct commodity shocks. Identifying this factor therefore provides a new way to recover historical changes in global economic activity and commodity prices that reflect endogenous responses to non-commodity-related shocks.

However, because standard empirical factor decompositions identify factors only up to a rotation, one cannot immediately recover the indirect common factor from a simple factor decomposition of commodity prices. The second element of our approach is to impose identification conditions, again grounded in the predictions of the theoretical model, to recover the direct and indirect factors underlying commodity price movements. The theoretical model provides two ways to do this: sign restrictions on factor loadings of the indirect common factor and orthogonality conditions with respect to a set of instruments for either the direct or indirect factors. Using a cross-section of 40 non-energy commodity prices available since 1968, we apply both identification strategies to identify the indirect factor and find similar results across specifications, indicating that the results are robust to the choice of identification strategy and instruments.

Our main empirical finding is that the vast majority of historical commodity price movements are associated with the indirect factor, i.e., broad-based changes in commodity prices can largely be attributed to a general-equilibrium response to aggregate non-commodity shocks rather than direct shocks to commodity markets. While there are a number of historical episodes during which direct shocks to commodity markets played some role in accounting for commodity price movements and changes in global production (e.g., 1979–80, the run-up in commodity prices in the 2000s and the decline in prices in 2008–09), the primary source of commodity price movements is their endogenous response to non-commodity-related shocks.

Our approach is related to the literature on the macroeconomic effects of shocks to oil and commodity prices (Bosworth and Lawrence 1982; Hamilton 1983; Barsky and Kilian 2002; Hamilton 2009; Blinder and Rudd 2012; Stuermer 2017) as well as a growing body of recent research on identifying the sources of oil price movements following Kilian (2009). We differ from this line of research in that we focus on a broad range of non-energy commodities (rather than just oil), which is essential to implement our identification strategy. Second, our identification strategy is new: we apply factor methods that decompose the comovement across different commodity prices then exploit the predictions about this decomposition from a microfounded model to identify the structural sources of fluctuations in commodity prices and aggregate output.¹ Third, while identification in VARs of commodity markets decomposes shocks into supply and demand shocks, our general-equilibrium model allows for the fact that exogenous forces should tend to have *both* supply and demand effects.

Our model provides a structural interpretation of a factor representation for commodity prices along with the requisite identification conditions, so that we are able to disentangle the different *economic* channels underlying commodity price movements. In this respect, our approach is related to work that uses economic theory to assign factors an economic interpretation (e.g. Forni and Reichlin 1998). Other work identifies the factors driving macroeconomic aggregates common to all countries and specific subsets of countries (Stock and Watson 2005; Kose et al., 2012). Factor methods have also been used to identify relative price changes for specific goods and the absolute price changes common to all goods (Reis and Watson 2010) and the relative importance of aggregate and sector-specific shocks for U.S. industrial production (Foerster et al., 2011). Our paper differs from this line of research in that we use commodity price dynamics to identify the sources of global

¹ We are not the first to apply factor methods to commodity prices. Some papers have examined whether there is excess comovement among unrelated commodities—that is, price comovement in excess of what one would expect, conditional on macroeconomic fundamentals (e.g. Pindyck and Rotemberg 1990). Other papers have investigated the forecasting performance of the common factor in metals prices for individual metals prices (West and Wong 2014) and commodity convenience yields for inflation (Gospodinov and Ng 2013). But there has been little attempt at interpreting the resulting factors in a structural sense.

business cycle fluctuations and in our identification strategy, which relies on the use of sign restrictions and orthogonality conditions rather than zero restrictions on the factor loadings.

Finally, our factor-based method can help with forecasting commodity prices. Using recursive out-of-sample forecasts, we find that a bivariate factor-augmented VAR with each commodity's price and the first common factor extracted from the cross-section of commodities generates improvements in forecast accuracy relative to the no-change forecast, particularly at short horizons. This result extends to broader commodity price indices as well as real oil prices. An additional advantage of our approach is that it relies only on commodity prices that can be readily updated at monthly or quarterly frequencies and does not require information about production and inventory data that are often unavailable at these frequencies. Our approach thus provides a unified framework to forecast commodity prices in real-time as well as a structural interpretation of these forecasts.

The remainder of the paper is organized as follows. Section 2 presents a general-equilibrium business cycle model with commodities and shows how the model can be used to assign a structural interpretation to the common factors in commodity prices. The section also shows how to recover the economic factors from typical factor decompositions through identification restrictions. Section 3 applies these results to a historical cross-section of commodity prices. Section 4 uses the indirect common factor in a recursive out-of-sample forecasting exercise. Section 5 concludes.

2. The sources of commodity price comovement: theory

In this section, we present a model that characterizes the sources of commodity price comovement. In particular, we show that the model yields a tractable factor structure for a cross-section of commodity prices, which permits an economic interpretation of the factors.

2.1. Model of commodity prices

The baseline model consists of households, an energy-producing sector, a continuum of heterogeneous primary commodities, a sector that aggregates these commodities into a single intermediate commodity input, and a final goods sector that combines commodities, labor and technology into a final good.

2.1.1. The household

A representative consumer maximizes expected discounted utility over consumption (C), labor supplied to the final good sector (N^S) and labor supplied to the energy sector (N^e):

$$\max E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{C_{t+i}^{1-\sigma}}{1-\sigma} - e^{-\varepsilon_{t+i}^n} \varphi_n \frac{N_{t+i}^S}{1 + \frac{1}{\eta}} - \varphi_e \frac{N_{t+i}^e}{1 + \frac{1}{\eta}} \right]$$

where β is the discount factor. We refer to the two types of labor as final-goods labor and energy-specific labor. With $\varphi_n > 0$ and $\varphi_e > 0$, welfare is decreasing in hours worked in either sector. The $e^{\varepsilon_t^n}$ term is an exogenous shock to the disutility of hours worked in the final goods sector.²

The household pays a price P_t for the consumption good, receives wage W_t for each unit of labor supplied to the final goods sector and wage W_t^e for labor in the energy sector. The household also can purchase risk-free bonds B_t that pay a gross nominal interest rate of R_t . The budget constraint is $P_t C_t + B_t = B_{t-1} R_{t-1} + W_t N_t^S + W_t^e N_t^e + T_t$ where T_t represents payments from owning firms.

2.1.2. The energy production sector

A representative energy firm produces a total supply of the energy good (Σ_t) subject to an exogenous energy-specific productivity shock (Ξ_t) and a decreasing returns production function that uses energy-specific labor ($N_t^{d,e}$): $\Sigma_t = \Xi_t (N_t^{d,e})^{1-\gamma}$. This representative firm takes the price of the energy good (S_t) as given and therefore chooses how much labor to hire to maximize its profits which are given by $S_t \Xi_t (N_t^{d,e})^{1-\gamma} - W_t^e N_t^{d,e}$. Equilibrium in the labor market for energy requires that $N_t^{d,e} = N_t^e$.

2.1.3. The primary commodity-production sector

There is a continuum of primary commodities of mass 1. Each primary commodity j is produced by a representative price-taking firm which uses energy $\Sigma_t(j)$ to produce an amount of the commodity $Q_t(j)$ using a decreasing returns in energy production function $Q_t(j) = A_t(j) \Sigma_t(j)^{1-\alpha_j}$ where $A_t(j)$ is the exogenously determined level of productivity for commodity j . The value $1 - \alpha_j$ ($0 < \alpha_j < 1$) determines the commodity-specific degree of diminishing returns to energy and is in equilibrium equal to the ratio of firm j 's expenditures on energy to its total revenues. Given the price of commodity j $P_t(j)$ and the price of energy S_t , the firm chooses the amount of energy input to maximize profits.

² One could alternatively model the household as providing a single type of labor which can be used in both the final goods and energy sector. This would not change the qualitative results. One could also introduce a preference shock to the household's willingness to supply labor to the energy sector but this would have the same qualitative effects as the productivity shock in the energy sector.

We assume the steady-state level of productivity $\bar{A}(j)$ is such that the steady-state level of production in each sector is equal. Equilibrium in the supply and demand for energy requires that $\Sigma_t = \int_0^1 \Sigma_t(j) dj$.

2.1.4. The intermediate commodity

A perfectly competitive sector purchases $Y_t(j)$ of each primary commodity j and aggregates it into an intermediate commodity Q_t^C using the Dixit-Stiglitz aggregator

$$Q_t^C = \left(\int_0^1 Y_t(j)^{\frac{\theta_c-1}{\theta_c}} dj \right)^{\frac{\theta_c}{\theta_c-1}}$$

where θ_c is the elasticity of substitution across commodities and the price of the intermediate commodity aggregate is given by $P_t^C = (\int_0^1 P_t(j)^{1-\theta_c} dj)^{\frac{1}{1-\theta_c}}$. Market clearing for each commodity sector j requires $Q_t(j) = Y_t(j)$. This setup implicitly assumes that no storage of commodities takes place, since all commodities produced must be used in the same period. We discuss the rationale for this assumption and its implications in more detail in Appendix I.

2.1.5. The final goods sector

A perfectly competitive sector combines purchases of the intermediate commodity good Y_t^C and labor N_t^d (at prices P_t^C and W_t respectively) according to the Cobb-Douglas production function $Y_t = A_t Y_t^{C\alpha_t} N_t^{d1-\alpha_t}$ to maximize profits, taking all prices as given and where A_t is an exogenously determined aggregate productivity process.³ Since all of the final good is purchased by the household, equilibrium in the final goods market requires $C_t = Y_t$ as well as $N_t^D = N_t^S$ and $Q_t^C = Y_t^C$. The fact that α_t may be time-varying allows for exogenous variation in the relative demand for commodities and labor in the production of the final good.

2.1.6. The linearized model

A detailed solution of the model is provided in Appendix A. We assume that exogenous processes are stationary around their steady-state levels, so that all real variables are constant in the steady state. Lower-case letters denote log deviations from steady state (e.g., $c_t \equiv \log C_t - \log \bar{C}$), and we normalize the nominal variables by the price level of final goods (e.g., $p_t(j) \equiv \log P_t(j)/P_t - \log(\bar{P}(j)/\bar{P})$). We normalize commodity-specific productivity as $v_t(j) \equiv a_t(j)(1 + (\varepsilon_j \theta_c)^{-1})^{-1}(1 + \varepsilon_j^{-1})$ to simplify the aggregation across commodities, where $\varepsilon_j \equiv (1 - \alpha_j)/\alpha_j$.⁴ We assume that the productivity process for each commodity sector has an idiosyncratic component and a common component such that $v_t(j) = v_t^a + v_t^j$, which implies that the sum of productivity across commodities is $v_t = \int_0^1 v_t(j) dj = v_t^a$. The idiosyncratic components are orthogonal across commodity sectors, such that $E[v_t^j v_t^k] = 0 \forall j \neq k$ and $E[v_t] = 0$. The log deviation of α_t from its steady-state value of α is denoted by $\check{\alpha}_t$ and similarly for $\check{\varepsilon}_t$. Each of the exogenous processes ($v_t, v_t^j, \check{\alpha}_t, \check{\varepsilon}_t, \varepsilon_t^n$) is stochastic, persistent, and assumed to be stationary, but we do not need to otherwise specify the specific process followed by each.⁵

As shown in Appendix 1, the aggregate level of production of final goods can be expressed in terms of exogenous forces:

$$y_t = \omega_y [a_t + \delta_n \varepsilon_t^n + \delta_{\varepsilon} \check{\varepsilon}_t + \delta_v v_t + \delta_{\alpha} \check{\alpha}_t] \quad (1)$$

where $\omega_y > 0$ (as long as $\sigma > 1$), $\delta_n > 0$, $\delta_{\varepsilon} > 0$ and $\delta_v > 0$: output rises with aggregate productivity as well as energy-specific and commodity-specific productivity shocks. Whether output rises when the relative demand for commodities increases ($\check{\alpha}_t$) depends on specific parameter values.

2.2. Comovement and the factor structure of commodity prices

As shown in Appendix 1, the supply of commodity j can be written in reduced form as:

$$p_t(j) = \varepsilon_j y_t(j) + \alpha_y y_t + \alpha_v v_t^j + \alpha_{\varepsilon} \check{\varepsilon}_t + \alpha_{\check{\alpha}} \check{\alpha}_t + \alpha_{vj} v_t^j. \quad (2)$$

Because different commodities have different returns to energy in the production process, the slope of their supply curves (given by ε_j) will generally differ. The supply curve of each commodity shifts up with increases in aggregate income y_t regardless of its source: greater aggregate production raises the price of energy and therefore the price of commodity j ($\alpha_y > 0$). The price of energy is higher when aggregate income rises because that higher income induces the household to reduce its supply of labor to the energy sector (which raises W_t^e and therefore the price of energy) and because higher

³ One could allow the final goods sector to also use energy in its production process without changing any of the qualitative results in the factor structure of commodity prices.

⁴ The rescaling of the commodity-specific productivity process ensures that a 1% increase in productivity in each commodity sector raises the equilibrium level of production of that commodity by equal amounts for each commodity. This would not be the case without the rescaling because each primary commodity sector's supply curve has a different slope.

⁵ Because we do not need to specify the specific time series process followed by each exogenous variable, we will frequently refer to these exogenous variables as "shocks" with obvious abuse of terminology, instead of referring explicitly to the innovations that generate variation in each exogenous variable.

aggregate income implies a greater aggregate demand for energy which also causes the price of energy to rise. Hence, any shock that affects aggregate income in this model is a supply shock from the point of view of a given commodity. In addition, productivity shocks in the commodity sector or productivity shocks to the energy sector will induce an additional shift of the supply curve, holding constant their effect on aggregate production, as will shocks to the relative demand for commodities (which affect energy prices and therefore change cost of producing commodities). Hence, these shocks have a direct effect on the supply of commodities, above and beyond the general equilibrium effects that all shocks have on supply via their effects on aggregate income.

The demand for commodity j can similarly be written in reduced form as:

$$p_t(j) = -\frac{1}{\theta_c} y_t(j) + \beta_y y_t + \beta_v v_t^a + \beta_\Xi \tilde{\Xi}_t + \beta_\alpha \tilde{\alpha}_t. \quad (3)$$

Given the setup of the model, all commodities have the same elasticity of demand. In addition, all commodities experience a rise in demand from an increase in aggregate production, regardless of its source ($\beta_y > 0$) which simply reflects the role of commodities as an input into the production of final goods. This term therefore captures general-equilibrium demand effects, and all macroeconomic shocks that affect aggregate production in the model result in an equal upward or downward shift in the demand for each commodity. Thus, all shocks in the model other than idiosyncratic shocks are both demand and supply shocks. However, in addition to these general-equilibrium shifts in commodity demand, the demand for commodity j rises with changes in the relative demand for commodities ($\tilde{\alpha}_t$), holding aggregate output constant. It also shifts, holding aggregate output constant, with exogenous common commodity productivity shocks (which increase the demand for all commodities on the part of the final goods sector) and exogenous shocks to energy (which affect the relative demand for all commodities).

As can be seen from the supply and demand curves above, there are two aggregate shocks that affect aggregate output but do not directly impact commodity prices other than through general equilibrium effects on output: a_t and ε_t^n . We will refer to these variables as “indirect” shocks that affect commodity prices only through their general equilibrium effects on output. It’s helpful to decompose movements in output coming from these variables vs other exogenous forces:

$$y_t = y_t^{nc}(a_t, \varepsilon_t^n) + \omega_y [\delta_\Xi \tilde{\Xi}_t + \delta_v v_t^a + \delta_\alpha \tilde{\alpha}_t]$$

where $y_t^{nc} = \omega_y[a_t + \delta_n \varepsilon_t^n]$ is the level of aggregate output coming exclusively from changes in aggregate productivity and changes in the willingness of households to supply labor to the final goods sector.

As shown in Appendix A, we can then rewrite the equilibrium price of commodity j as

$$\begin{aligned} p_t(j) &= \underbrace{\lambda_j^y y_t^{nc}(a_t, \varepsilon_t^n)}_{\text{indirect (IC)}} + \underbrace{\lambda_j^\Xi \tilde{\Xi}_t + \lambda_j^v v_t^a + \lambda_j^\alpha \tilde{\alpha}_t}_{\text{direct (DC)}} - \underbrace{\frac{1}{\theta_c} v_t^j(j)}_{\text{idiosyncratic}} \\ &= \lambda_j F_t + \xi_t^j \end{aligned} \quad (4)$$

where $\lambda_j^y > 0$ and $\frac{\partial \lambda_j^\Xi}{\partial (1-\alpha_j)} < 0$ when $\sigma > 1$ and $\theta_c(1-\alpha) - 1 > 0$.

Eq. (4) provides a factor structure for real commodity prices with three distinct and orthogonal components.⁶ The last term on the right-hand side reflects idiosyncratic shocks to commodity j that have no aggregate real effects. The second set of terms on the right-hand side consist of a factor for each exogenous force that has both direct and indirect effects on the commodity market (i.e., that shifts the supply or demand for commodities, holding aggregate output constant, but also ultimately leads to changes in aggregate output). For this reason, we refer to these factors as “direct common” (DC) factors. In this setup, there are three such factors: exogenous shocks to energy sector, a common productivity shock to commodities, and a shock to the relative demand for commodities in the production of final goods. Because these forces have *both* direct and indirect effects on the market for commodity j , there is, in general, no guarantee that their respective loadings have the same signs across commodities. However, it should be the case that commodities which are more intensive in energy (larger α_j) should have prices which respond more strongly to exogenous energy price shocks.

The most interesting component of the factor structure is the first term on the right-hand side of (4), which reflects the *combined* contribution on the price of commodity j from all shocks whose effects on commodity prices operate only indirectly through aggregate output (i.e., only through general-equilibrium effects). We refer to this common factor as the “indirect common” (IC) factor. It captures the fact that, because some shocks affect commodity markets only through changes in aggregate output, they all have identical implications for the price of a given commodity, conditional on the size of their effect on aggregate output, and induce the same comovement across different commodity prices. As a result, they can be represented as a single factor. Furthermore, this factor has a well-defined interpretation: it is the counterfactual level of

⁶ The differences in factor loadings across commodities in Eq. (4) stem only from differences in slopes of commodity supply curves, which are themselves a reflection of the different energy intensities of production across commodities. Appendix 2 considers a version of the model where industrial and agricultural commodities are aggregated into two different intermediate commodity bundles, both of which are used in final production, but in which the aggregation occurs with different elasticities of substitution. The factor structure is preserved, even though commodities now differ in both the slopes of their supply curves as well as their demand curves. This alternative representation of the model also illustrates that the aggregation of shocks underlying the factor structure of commodity prices does not hinge on the use of CES aggregation of all commodities into a single commodity intermediate input.

global output that would have occurred in the absence of any direct commodity shocks. Thus, this common factor represents a way to reconstruct the counterfactual history of aggregate output without direct commodity shocks, as well as to decompose historical commodity price changes into those components reflecting direct commodity shocks versus all other aggregate economic forces captured by the IC factor. Unlike the DC factors, another key characteristic of the IC factor is that all the loadings on this factor must be positive ($\lambda_j^y > 0 \forall j$). This prediction reflects the fact that the shocks incorporated in the IC factor raise commodity demand when the shock is expansionary and simultaneously restrict the commodity supply through income effects, which unambiguously increases commodity prices.⁷ In short, this factor decomposition provides a new way to separate causality in the presence of simultaneously determined prices and production levels.

2.3. Recovering a structural interpretation of the factors

A key limitation of factor structures is that, empirically, factors are identified only up to a rotation. For example, if one estimated a factor structure on commodity prices, one could not directly associate the extracted factors with the structural interpretation suggested by (4). However, the theory developed in this section has implications that can be used to identify the unique rotation consistent with those predictions and permits us to assign an economic interpretation to the factors driving commodity prices.

To see this, suppose that, as in the theory above, the N variables in vector X_t (N by 1) of real commodity prices have a factor structure $X_t = LF_t + \varepsilon_t$ where F_t is a K by 1 vector of unobserved variables, and L is an N by K matrix of factor loadings. Let the variance of ε_i be given by φ_i and the covariance matrix of ε_i be $\text{cov}(\varepsilon) = \text{diag}(\varphi_i) = \Psi$ such that the ε_i 's are uncorrelated with one another. We make the typical assumptions underlying factor analysis: (a) $E(F) = 0$, (b) $E(\varepsilon_i) = 0$, (c) $E(F\varepsilon_i) = 0$ and (d) $\text{cov}(F) = I$, so that the factors are orthogonal to one another and have variance normalized to one. Then, letting $\Sigma \equiv \text{cov}(X)$ be the covariance matrix of X , it follows that $\Sigma = LL' + \Psi$. The identification problem is that for any K by K orthogonal matrix T such that $TT' = I$, we can define $\tilde{L} = LT$ and $\tilde{F}_t = T'F_t$ such that $X_t = \tilde{L}\tilde{F}_t + \varepsilon_t$. As a result, an empirical estimate of the factors underlying X_t does not, in general, permit the economic identification of the factors F_t but rather some rotation $\tilde{F}_t$.

However, the model provides additional restrictions on the factor structure that can be used to assign an economic interpretation to the factors and identify the “structural” factors F_t from the estimated factors $\tilde{F}_t$. For example, consider the factor structure of Eq. (4) in Section 2.2 in which real commodity prices reflect two underlying factors, an exogenous energy shock ($\tilde{\varepsilon}_t$) and the level of aggregate production that would have occurred in the absence of this shock (y_t^{nc}), thus $F_t = [F_t^1 \ F_t^2]' = [y_t^{nc} \ \tilde{\varepsilon}_t]'$. As we discuss below, this two-factor structure is the most empirically relevant case. A factor decomposition of commodity prices would yield a rotation of these factors $\tilde{F}_t$ such that

$$F_t = T'\tilde{F}_t = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix}' [\tilde{F}_t^1 \ \tilde{F}_t^2]' = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} [\tilde{F}_t^1 \ \tilde{F}_t^2]'$$

where the last equality reflects the properties of rotation matrices. Recovering the “structural” factors F_t corresponds to identifying the parameter θ and the rotation matrix T such that $F_t = T'\tilde{F}_t$.

The theory imposes three types of conditions that can be used to identify θ . The first is that y_t^{nc} (the IC factor) is orthogonal to commodity-related shocks (DC factors). Therefore, if one had a S by 1 vector of instruments z_t that is correlated with the energy shocks $\tilde{\varepsilon}_t$, the orthogonality of y_t^{nc} would deliver S moment conditions $E[y_t^{nc}z_t] = 0$. The conditions can be rewritten as

$$E[y_t^{nc}z_t] = E[(\tilde{F}_t^1 \cos \theta + \tilde{F}_t^2 \sin \theta)z_t] = 0. \quad (5)$$

If $S = 1$, then θ would be uniquely identified. If $S > 1$, then θ is overidentified, and one could estimate it using generalized method of moments (GMM) by writing the moment conditions as

$$J(\theta) = E[y_t^{nc}z_t]WE[y_t^{nc}z_t]' \quad (6)$$

where W is a weighting matrix, such that $\hat{\theta} = \text{argmin } J(\theta)$. Letting W be the inverse of the variance-covariance matrix associated with the moment conditions, standard GMM asymptotic results apply, including standard errors for θ and tests of the over-identifying conditions for N and T large enough for the factors to be considered as observed variables rather than generated (e.g., Stock and Watson 2002; Bai and Ng 2002).

A second approach would be to make use of the theoretical prediction that y_t^{nc} is a linear combination of exogenous variables that have only indirect effects on the commodity sector such as the productivity shocks or labor supply shocks considered in the model. If one had a S by 1 vector of instruments z_t for each period correlated with one or more of these exogenous drivers, then another set of orthogonality conditions imposed by the theory would be $E[F_t^2z_t] = 0$. As in the previous case, one could estimate θ using GMM, given these orthogonality conditions, and test over-identifying restrictions if $S > 1$.

⁷ The prediction that all commodities have the same sign loading on the common factor is sensitive to the assumption of diminishing returns to scale in production of commodities, which generates upward-sloping supply curves. Downward sloping supply curves (coming from increasing returns) would imply that when global production is high, the increased demand would now reduce commodity prices (since supply slopes down), while the income effect would raise it (as before). Whether commodity prices went up would therefore depend on the relative strengths and slopes of each effect.

In both of these cases, the econometrician must take a stand on whether the chosen instruments should be correlated with commodity-related shocks or with y_t^{nc} . While economic theory may provide clear guidance in some cases, this choice may be problematic when one is interested in whether an exogenous variable affects commodities only through general-equilibrium effects or more directly. Within our framework, this question amounts to whether the exogenous variable should be considered part of y_t^{nc} or one of the commodity-related shocks. For example, in the case of commodity prices, monetary policy shocks could potentially have direct effects on commodity markets in the presence of storage motives but would otherwise not be expected to have direct effects on commodity markets if the speculative channel is absent or sufficiently small, as discussed in Appendix I.

A third approach is to make use of sign restrictions on the loadings. The theory predicts that the loadings on y_t^{nc} must all be positive (since $\lambda_j^y > 0 \forall j$ in Eq. (4)). Letting $\tilde{L}$ be the N by 2 matrix of unrotated factor loadings, the rotated or “structural” loadings are $L = \tilde{L}T = [\tilde{L}^1 \ \tilde{L}^2]T$. The loadings on the first rotated factor (corresponding to y_t^{nc}) are then $L^1 = \tilde{L}^1 \cos \theta + \tilde{L}^2 \sin \theta$. Imposing that all of the elements of L^1 be positive would therefore correspond to identifying the range of values of θ such that $\min(\tilde{L}^1 \cos \theta + \tilde{L}^2 \sin \theta) > 0$. In general, this leads only to a set of admissible values of θ and associated rotation matrices without uniquely identifying the rotation matrix. This approach is conceptually similar to the set identification of VARs by sign restrictions (Uhlig, 2002).

In short, the theoretical model of commodity prices yields not only a factor structure for commodity prices but also a set of conditions that can be used to identify (or, in the case of sign restrictions, limit the set of) the rotation matrix necessary to recover the underlying factors. Furthermore, the factors have economic interpretations. The IC factor corresponds to the level of production and income net of commodity-related shocks, while other factors correspond to one or more of these commodity-related shocks. The identification of the rotation matrix, and thus the underlying economic factors, follows from orthogonality conditions implied by the model, as well as sign restrictions on the loadings predicted by the theory. The implied factor structure of the model combined with the ability to recover an economic interpretation of the factors thus provides a new method for separating fluctuations in aggregate output into those driven by commodity-related shocks and those driven by non-commodity-related shocks.

3. The sources of commodity price comovement: empirical evidence

In this section, we implement the factor decomposition of real commodity prices suggested by the theory. We first construct a historical cross-section of real commodity prices for the commodities that conform to the theoretical structure of the model along several dimensions. We then implement a factor decomposition and identify the factors suggested by the theory. After considering a wide range of robustness checks, we argue that commodity-related shocks have contributed only modestly to fluctuations in global economic activity.

3.1. Data

Guided by the theoretical model, we use four criteria to decide which commodities to include in the data set and which to exclude. First, commodities must not be vertically integrated.⁸ Second, the main use of commodities must be directly related to the aggregate consumption bundle, and they should not be primarily used for the purposes of financial speculation.⁹ Third, commodities must not be jointly produced.¹⁰ Finally, the pricing of commodities must be determined freely in spot markets and must not display the price stickiness associated with the existence of long-term contractual agreements.¹¹

Applying these criteria leaves us with 40 commodities in the sample. It includes 22 commodities that we refer to as agricultural or food commodities, five food oils, and 13 industrial commodities (see Appendix C for a detailed list). We compiled monthly data from January 1957 to January 2013 (as available) from a number of sources, including the CRB Commodity Yearbooks, the CRB InfoTech CD, the World Bank GEM Commodity Price Data, the IMF's Commodity Price Indices and the U.S. Bureau of Labor Statistics. While most of the data are consistently available from January 1968 until January 2013, in some cases, there are a number of missing observations in the underlying data, as well as periods when we treat the

⁸ Vertically integrated commodities would introduce the possibility of price comovement resulting from idiosyncratic shocks to one commodity affecting prices in other commodities through the supply chain. For example, an exogenous shock to the production of sorghum would affect the price of non-grass-fed beef because sorghum is primarily used as feed. Thus, this shock could ultimately affect the price of milk and hides as well.

⁹ Some commodities, such as precious metals, have long been recognized as behaving more like financial assets than normal commodities (Chinn and Coibion 2014). Thus, we exclude gold, silver, platinum and palladium from the cross-section of commodities.

¹⁰ Some commodities are derivative products of the production of other commodities. This is particularly the case for minerals, which are commonly recovered during the mining for metal commodities, making the assumption of orthogonal productivity shocks clearly inapplicable. We drop any such commodities.

¹¹ While many commodities have long been traded on liquid international spot markets, this is not always the case. For example, the price measure of tung oil tracked by the Commodity Research Bureau (CRB) Commodity Yearbooks varies little over time and is often fixed for periods lasting as long as one year. Because we want to focus on commodities whose prices reflect contemporaneous economic conditions, we exclude commodities such as tung oil that systematically display long periods of price invariance. For some commodities in the sample, prices were not determined in flexible markets until much later than others; for these commodities, we treat early price data as missing values. For mercury, the reverse is true, since its use has declined over time and its price began to display long periods with no price changes starting in 1995. We treat its prices after March 1995 as missing. Appendix B provides more details on these adjustments.

Table 1
Contribution of common factors to commodity prices.

Number of Common Factors:	Cumulative variance explained by common factors				
	1	2	3	4	5
<i>Complete Sample:</i>					
Cumulative eigenvalue shares	0.59	0.69	0.75	0.79	0.82
Mean across commodity-specific R^2 s	0.60	0.69	0.74	0.78	0.81
Median across commodity-specific R^2 s	0.70	0.76	0.78	0.84	0.85
R^2 across all commodities	0.62	0.71	0.75	0.79	0.82
<i>Subset of Commodities:</i>					
R^2 across agricultural/food commodities	0.64	0.72	0.75	0.77	0.80
R^2 across oils	0.72	0.74	0.76	0.82	0.85
R^2 across industrial commodities	0.55	0.68	0.75	0.80	0.83

Notes: The table provides metrics of the cumulative variance associated with using additional factors, as indicated by each column. The first row provides the cumulative sum of eigenvalues associated with each factor normalized by the sum of all eigenvalues. The second row provides the mean across the R^2 of each commodity for each given factor, using the specific sample associated with each commodity. The third row provides the median R^2 across all commodity-specific R^2 s. The fourth row provides the joint R^2 constructed using all commodities. In addition, the top panel presents joint R^2 s for subsets of commodities (as defined in Table 1). Each R^2 omits imputed values. See Section 3.2 for details.

available data as missing because spot trading was limited. Appendix C provides details on the construction of each series, their availability and any periods over which we treat the data as missing because of infrequent price changes. Furthermore, while we can construct price data going back to at least 1957 for many commodities, we restrict the empirical analysis to the period since 1968, in light of the numerous price regulations and government price support mechanisms in place during the earlier period.¹²

3.2. Common factors in commodity prices

Before conducting the factor analysis, we normalize each price series by the U.S. CPI, take logs of all series and normalize each series by its standard deviation. Because there are missing observations in the data, we use the expectation-maximization (EM) algorithm of Stock and Watson (2002).¹³ We follow Kilian (2009) in focusing on the (log) level of real commodity prices but document in our robustness checks that our results are qualitatively unchanged if we take the first difference of real commodity prices or use linearly detrended series.

We consider several metrics to characterize the contribution of the first five factors in accounting for commodity price movements, summarized in Table 1.¹⁴ The first row presents the sum of eigenvalues associated with each number of factors normalized by the sum across all eigenvalues, a simple measure of variance explained by common factors. In addition, we present additional metrics based on R^2 s that explicitly take into account missing values associated with some commodities: the average across the individual R^2 s computed for each commodity, the median across these same commodity-specific R^2 s, the R^2 constructed across all commodities. The key result from this table is that the first common factor explains a large share of the price variation across commodities, ranging from 60% to 70% depending on the specific measure used. In contrast, all of the additional factors explain smaller percentages of the variance in commodity prices. The second factor, for example, accounts for between 6% and 10%, while the third factor contributes another 5% of the variance. Thus, the first two factors jointly account for approximately 70–75% of the variance in commodity prices.¹⁵

The ability of the first two factors, and the first common factor in particular, to account for so much of the variance holds across commodity groups. Table 1 includes the contribution of different factors to explaining the variance across the three subsets of commodities in the sample—agricultural/food, oils and industrials. Differences across subsets of commodities are quite small: the contribution of the first factor ranges from 55% (pooled R^2 across all commodities in this subset) for industrial commodities to 64% for agricultural/food commodities and 72% for oils (see Appendix E for commodity-specific results). The decomposition does not suggest that one needs different factors for different types of commodities. This point is worth stressing because a common concern with factor analysis is that different factors are needed to explain different subsets of the data. As illustrated in Table 1, this is not the case for these commodities.

¹² Appendix D provides detail on the geographic variation in where commodities are produced and how they are used.

¹³ Specifically, we first demean each series and replace missing values with zeroes before recovering the first K factors. We use these K factors to impute the value of missing observations, and then do the factor analysis again, iterating on this procedure until convergence. We use $K=5$ factors for the imputation; however, the results are not sensitive to the specific number of factors used.

¹⁴ Following Connor and Korajczyk (1993) and Bai and Ng (2002), we use principal components on the variance-covariance matrix of commodity prices to estimate the approximate factors.

¹⁵ Statistical tests of the number of factors point toward parsimonious factor specifications. For example, the PC2 and IC2 criteria of Bai and Ng (2002) each select one factor. The same result is obtained using the test suggested by Onatski (2010) or the two criteria proposed in Ahn and Horenstein (2013).

Table 2
GMM estimates of the rotation matrix.

	GMM estimates of rotation parameter				Implied rotation coefficients			
	θ	$se(\theta)$	$p(\text{over-id})$	N	t_{11}	95% CI(t_{11})	t_{21}	95% CI(t_{21})
Baseline GMM Estimates: (Iterative GMM, $L=36$)	−0.10	(0.31)	1.00	505	1.00	[0.75 1.00]	−0.10	[−0.65 0.49]
Robustness of GMM Estimates:								
More moments: ($L=48$)	−0.15	(0.27)	1.00	493	0.99	[0.77 1.00]	−0.15	[−0.63 0.39]
Fewer moments: ($L=24$)	−0.13	(0.35)	1.00	517	0.99	[0.67 1.00]	−0.13	[−0.73 0.54]
Fewer moments: ($L=12$)	−0.23	(0.50)	1.00	529	0.97	[0.32 1.00]	−0.23	[−0.94 0.69]
Two-step GMM	−0.10	(0.31)	1.00	505	1.00	[0.75 1.00]	−0.10	[−0.65 0.47]
Continuous GMM	−0.07	(0.31)	1.00	505	1.00	[0.76 1.00]	−0.07	[−0.62 0.52]
Alternative normalization	−0.08	(0.31)	1.00	505	1.00	[0.75 1.00]	−0.08	[−0.64 0.50]

Notes: The table presents nonlinear GMM estimates of parameter θ from Eq. (7) in the text, along with Newey and West (1987) standard errors ($se(\theta)$), the p -value for over-identifying restrictions ($p(\text{over-id})$), and the number of observations used in the estimation (N). The panel on the right presents the implied parameters of the first row of the rotation matrix, along with the 95% confidence interval implied from the estimated distribution of θ . The baseline estimates are based on iterative GMM until convergence, using a constant as well as the contemporaneous value and 36 lags of OPEC production shocks for moment conditions. Subsequent rows present robustness to using more or fewer lags of OPEC production shocks as moment conditions, a two-step GMM procedure, a continuously updated GMM procedure and an alternative normalization of moment conditions. See Section 3.3 for details.

3.3. Identification of the rotation matrix and the underlying economic factors

To implement a structural interpretation of the factors as suggested by the model, we interpret the results of Table 1 as indicating that a two-factor representation adequately characterizes the data. First, additional factors beyond the first two add relatively little in explanatory power and can be omitted. Second, under the null of the model, it is *a priori* unlikely for there to be fewer than two factors. Indeed, such a finding would imply that there are *no* shocks that directly affect commodity prices and that all movements in commodity prices reflect either the level of aggregate economic activity or idiosyncratic commodity factors. We can rule this possibility out immediately because there exists at least one common shock to the supply of commodities: exogenous energy price movements. Because commodities require energy in production and distribution, exogenous shocks to energy prices necessarily induce some comovement in commodity prices, as commodities are produced in different parts of the world but consumption occurs disproportionately in advanced economies, thereby generating significant shipping and distribution costs.

To estimate the rotation matrix, our baseline is to impose orthogonality conditions on the indirect common factor F_t^1 . Specifically, we take $\varepsilon_t^{\text{OPEC}}$, the measure of OPEC production shocks from Bastianin and Manera's (2014) updated version of the Kilian (2008) series and define the orthogonality conditions as $E[F_t^1 z_t]$, where $z_t \equiv [1 \ \varepsilon_t^{\text{OPEC}} \ \dots \ \varepsilon_{t-L}^{\text{OPEC}}]$ is the vector of instruments that consists of a constant, the contemporaneous value of the production shock series as well as L lags of the shock. The IC factor F_t^1 (y_t^{nc} in the model) is a rotation over the two estimated factors $\hat{F}_t^1$ and $\hat{F}_t^2$, i.e., $F_t^1 = t_{11}\hat{F}_t^1 + t_{21}\hat{F}_t^2$ where the orthogonal rotation parameters t_{11} and t_{21} can be expressed as a function of a single underlying rotation parameter θ such that $t_{11} = \cos \theta$ and $t_{21} = \sin \theta$. Given that there are more moment conditions ($L+2$) than parameters (θ), we can estimate the rotation parameter θ using GMM by minimizing $J(\theta)$:

$$J(\theta) = \left[\frac{1}{T} \sum_t (F_t^1(\theta) z_t) \right] W \left[\frac{1}{T} \sum_t (F_t^1(\theta) z_t) \right]' \quad (7)$$

We set $L=36$ months for the baseline estimation to capture the fact that the OPEC production shocks have long-lived effects on commodity prices, although the results are robust to both shorter and longer lag specifications as well, as we document below. W is the Newey-West (1987) heteroskedasticity and autocorrelation HAC robust estimate of the inverse of the variance-covariance matrix of moment conditions. We iterate over minimizing $J(\theta)$ and then computing the implied weighting matrix until the estimate of θ has converged ($W=I$ in the first step). Table 2 presents the resulting estimate of θ and its associated standard error. With $\hat{\theta} = -0.10$ and a standard error of 0.31, we cannot reject the null hypothesis that $\theta=0$. From this estimate of θ , we construct estimates of the rotation parameters t_{11} and t_{21} : t_{11} is close to 1, and we cannot reject the null hypothesis that $t_{21} = 0$, so the estimated rotation matrix is not statistically different from the identity matrix. Furthermore, the over-identification conditions cannot be rejected.

The reason why the estimated rotation matrix is close to the identity matrix is that, while the first unrotated factor is largely uncorrelated with OPEC production shocks, this condition is not satisfied for the second unrotated factor, which responds strongly to OPEC production shocks. Because the unrotated factors are already largely consistent with the theoretically predicted orthogonality conditions (namely, that the first factor is orthogonal to oil shocks, but the second is not), the estimation procedure yields only a slight rotation of the original factors.

While the fact that we cannot reject the over-identifying conditions is consistent with the theory, we can further assess the extent to which the estimated rotation satisfies the theoretical predictions of the model. For example, an additional theoretical prediction is that the loadings on the indirect factor will all be the same sign. To assess this prediction, we

Table 3
Rotated commodity-specific factor loadings.

Commodity	Factor loadings		Commodity	Factor loadings	
	IC	DC		IC	DC
<i>Agr./Food Commodities</i>			<i>Oils</i>		
Apples	0.46	0.13	Coconut oil	0.82	0.02
Bananas	0.57	0.22	Groundnut oil	0.86	0.13
Barley	0.75	0.41	Palm oil	0.89	0.13
Beef	0.87	−0.09	Rapeseed oil	0.53	0.39
Cocoa	0.89	−0.12	Sun/Safflower oil	0.83	0.22
Coffee	0.85	−0.17			
Corn	0.95	0.09	<i>Industrial Commodities</i>		
Fishmeal	0.91	0.15	Aluminum	0.80	0.05
Hay	0.86	−0.04	Burlap	0.85	−0.00
Oats	0.88	0.11	Cement	0.21	0.06
Orange juice	0.74	−0.22	Copper	0.60	0.69
Onions	0.53	−0.39	Cotton	0.92	−0.20
Pepper	0.56	−0.62	Lead	0.73	0.58
Potatoes	0.73	−0.05	Lumber	0.53	−0.23
Rice	0.93	0.09	Mercury	0.46	0.75
Shrimp	0.44	−0.75	Nickel	0.20	0.74
Sorghums	0.95	0.08	Rubber	0.79	0.45
Soybeans	0.95	0.02	Tin	0.90	0.18
Sugar	0.78	0.11	Wool	0.87	0.16
Tea	0.87	−0.22	Zinc	0.60	0.36
Tobacco	0.84	−0.33			
Wheat	0.92	0.13			

Notes: The table presents the rotated loadings from factor analysis using the GMM estimates of the rotation matrix. See Section 3.3 for details.

present in Table 3 the estimated factor loadings for each rotated factor. The loadings on the IC factor are positive for all commodities, as predicted by the theory. In contrast, the loadings on the commodity-related factor are of mixed signs. There are no systematic patterns across commodity groups, again confirming that the factors explaining commodity prices are common across commodity subsets. Without imposing any restrictions on the loadings as part of the identification strategy for the rotation matrix, we find that the estimated rotation satisfies theoretical predictions on the factor loadings and those implied by the over-identifying restrictions.¹⁶

Given the estimate of θ and the rotation matrix, we construct the rotated factor F_t^1 that, according to the model, corresponds to the level of aggregate output and income that would have occurred in the absence of commodity-related shocks. This factor is presented in Fig. 1, after being detrended with a Hodrick-Prescott (HP) filter with $\lambda = 129,600$, the typical value for monthly data, to highlight variation at business cycle frequencies. In addition, we draw from the estimated distribution of θ , construct F_t^1 for each new draw and use this distribution to characterize the 99% confidence interval of the HP-filtered factor.

This factor displays a sharp rise in 1973–74 before falling sharply during the 1974–75 recession in the United States. This drop is followed by a progressive increase over the course of the mid- to late 1970s, with the factor peaking in 1979 before falling sharply during each of the “twin” recessions of 1980–82, and then rebounding sharply after the end of the Volcker disinflation. Thus, over the course of the 1970s, this structural factor displays a clear cyclical pattern. During the mid-1980s, the factor drops sharply before rebounding in the late 1980s, and then falls gradually through the 1990–91 U.S. recession before rebounding through the mid-1990s. It experiences a large decline in the late 1990s, before the 2000–01 U.S. recession and then rebounds shortly thereafter. After a brief decline in the mid-2000s, the factor displays a sharp increase from 2005 to 2008, the period when many commodity prices boomed, and then falls sharply in late 2008 and 2009 before rebounding strongly in 2010. In short, there is a clear procyclical pattern to the IC factor relative to U.S. economic conditions.

3.4. Robustness of the estimation

To assess the sensitivity of our results, we vary some of the specific choices made for the estimation of θ . For example, we report in Table 2 results from using fewer moment conditions ($L = 12$ and 24 months) as well as more moment conditions ($L = 48$ months). Neither changes the estimates significantly. Similarly, we repeat the GMM estimates using a two-step

¹⁶ The model developed in Section 2 also predicts that commodities that are more energy intensive in production have larger (in absolute value) loadings on the DC factor. Using the 2007 benchmark input-output tables to characterize energy intensity of production for 13 commodity groupings (which jointly include 34 of the commodities in our cross-section), we find a positive relationship between the loadings and the energy intensity of production. These results are reported in Appendix K. This suggestive evidence is consistent with the model's predictions and lends credibility to the framework we use to characterize the common variation in commodity prices. Unfortunately, the input tables are not sufficiently detailed to allow us to test this prediction for a more differentiated set of commodities.

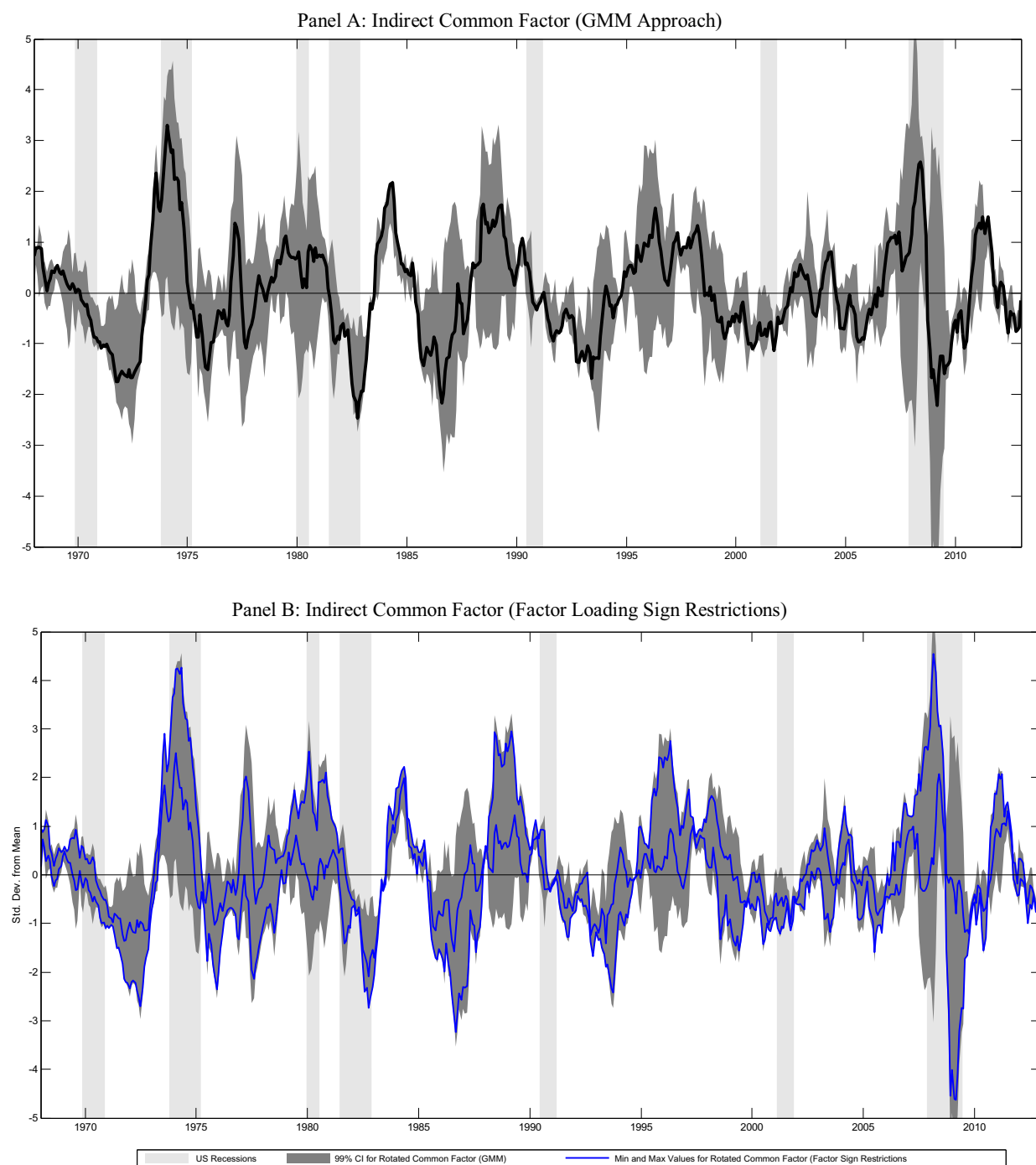


Fig. 1. Indirect common factor in commodity prices.

Notes: The figure in Panel A presents the IC factor from the factor analysis in Section 3.3. The IC factor is HP-filtered ($\lambda = 129,600$) in the figure. The light grey shaded areas are recessions dated by the National Bureau of Economic Research. The dark grey shaded areas are 99% confidence intervals of HP-filtered rotated factors constructed from the estimated distribution of rotation parameters. The figure in Panel B plots the 99% confidence interval of the IC factor as estimated by GMM (shaded areas), and the minimum and maximum range for admissible values of the IC factor using sign restrictions on factor loadings (solid blue lines). See Section 3.3 for details. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

procedure, in which θ is first estimated using a weighting matrix equal to the identity matrix with no subsequent iterations after updating the weighting matrix, and second using a continuously updated GMM in which we minimize over θ and W jointly until convergence. In both cases, the results are qualitatively similar. Finally, because non-linear GMM can be sensitive to normalizations, we replicate the baseline estimation after rewriting moment conditions as $E[(\hat{F}_t^1 + \hat{F}_t^2 \frac{\sin \theta}{\cos \theta})z_t] = 0$, and the results are again unchanged.¹⁷

A more interesting robustness check is to consider the alternative identification strategy suggested in section 2.3, namely to exploit the theoretical predictions for signs of factor loadings: loadings on the IC factor should all be positive. Thus, one can characterize the set of admissible rotation matrices by restricting them to be consistent with the sign restrictions implied by the theory, in the spirit of Uhlig (2002). In our case, this procedure consists of identifying the set of θ such that $\min(\tilde{L}^1 \cos \theta + \tilde{L}^2 \sin \theta) > 0$, where $\tilde{L}^i$ for $i = \{1, 2\}$ are the loading vectors associated with the unrotated factors and $\min$ is with respect to the elements of $\tilde{L}^1$. We consider values of $\theta \in [-\pi, \pi]$ (at increments of 0.001) and, for each θ , determine whether the restriction is satisfied. This yields a set of admissible rotation matrices and therefore a set of possible IC factors. We apply the HP filter to each of these and plot the resulting minimum and maximum values for each month in Panel B of Fig. 1, along with the 99% confidence interval for the rotated IC factor from the baseline GMM estimation. There is significant overlap between the two approaches, with the minimum and maximum values from the sign restriction typically being within the 99% confidence interval of the GMM-estimated IC factor. Thus, despite the fact that the two identification strategies are quite different, they point toward a remarkably consistent characterization of the non-commodity-related structural factor.

In Appendix F, we provide a number of additional robustness checks on these results. For example, we replicate our results dropping either all commodities whose primary use is as food or as feed, or those commodities disproportionately produced in the U.S.S.R., China or India through much of the sample. We verify that our results are robust to alternative assumptions about stationarity and find similar results using first differences or linearly detrended real commodity prices (Appendix G). We also find little sensitivity to dropping commodities for which imputation was needed or to decomposing the correlation matrix rather than the covariance matrix for the factor analysis.

3.5. Factor contributions to commodity prices, comovement and global real activity

We now turn to using the factor structure to better understand the historical sources of commodity price movements and global real activity. For prices, we decompose the average annual percentage change in commodity prices into the components driven by indirect and direct common factors. The decomposition follows directly from the rotated factor structure, yielding

$$\overline{p_t - p_{t-12}} = \overline{\lambda^{IC}}(F_t^{IC} - F_{t-12}^{IC}) + \overline{\lambda^{DC}}(F_t^{DC} - F_{t-12}^{DC}) + (\overline{\varepsilon_t} - \overline{\varepsilon_{t-12}})$$

where the bar denotes averages across all commodities in the cross-section. The first term on the right-hand side of the equation represents the contribution of the IC factor to average commodity price changes, the second represents the contribution of the DC factor, and the third reflects average idiosyncratic effects. The results of this decomposition are presented in the top panel of Fig. 2, in which we plot the contributions from the IC and DC factors each month as well as the actual annual average price change across commodities. The IC factor, which captures the endogenous response of prices to non-commodity shocks, explains the vast majority of historical commodity price changes. To the extent that income effects on inputs into the production of commodities are likely weak, the IC factor can be interpreted as primarily reflecting changing demand for commodities due to changes in global economic activity. During the commodity boom of 1973–74, for example, indirect shocks to commodity markets accounted for almost all of the rise in commodity prices, with the remainder reflecting direct commodity-related shocks. Every other historical episode of large changes in average commodity prices is also accounted for by the indirect factor, i.e., as an endogenous response of commodity prices to global business cycle conditions not driven by commodity-related shocks.

We can also quantify how changes in each factor have contributed to the time variation in *comovement* among commodity prices. Specifically, we can decompose, each month, annual changes in real commodity prices as follows:

$$p_t(j) - p_{t-12}(j) = \lambda_j^{IC}[F_t^{IC} - F_{t-12}^{IC}] + \lambda_j^{DC}[F_t^{DC} - F_{t-12}^{DC}] + \varepsilon_t^j - \varepsilon_{t-12}^j$$

From this, we can construct each month the cross-sectional R^2 coming from both factors (i.e., the ability of changes in both factors to explain commodity price movements through common forces) as well as the partial R^2 coming from the IC factor. These series are plotted in Panel B of Fig. 2. There is significant variation over time in the overall comovement of commodity prices, as captured by both factors, with the highest degrees of comovement in commodity prices occurring between 1973 and 1975, in the early to mid-1980s, in the late 1990s, and in the mid- to late 2000s continuing to 2013. The time variation in comovement is again primarily explained by changes in the indirect factor. Periods in which commodity prices co-move most strongly have also been periods in which commodity price changes have been driven by the endogenous response of commodity prices to non-commodity shocks.

¹⁷ While we rely on standard asymptotics which apply for large N and T , adjusting standard errors to account for smaller N would only strengthen the main result that one cannot reject the null that the estimated rotation matrix is not different from the identity matrix.

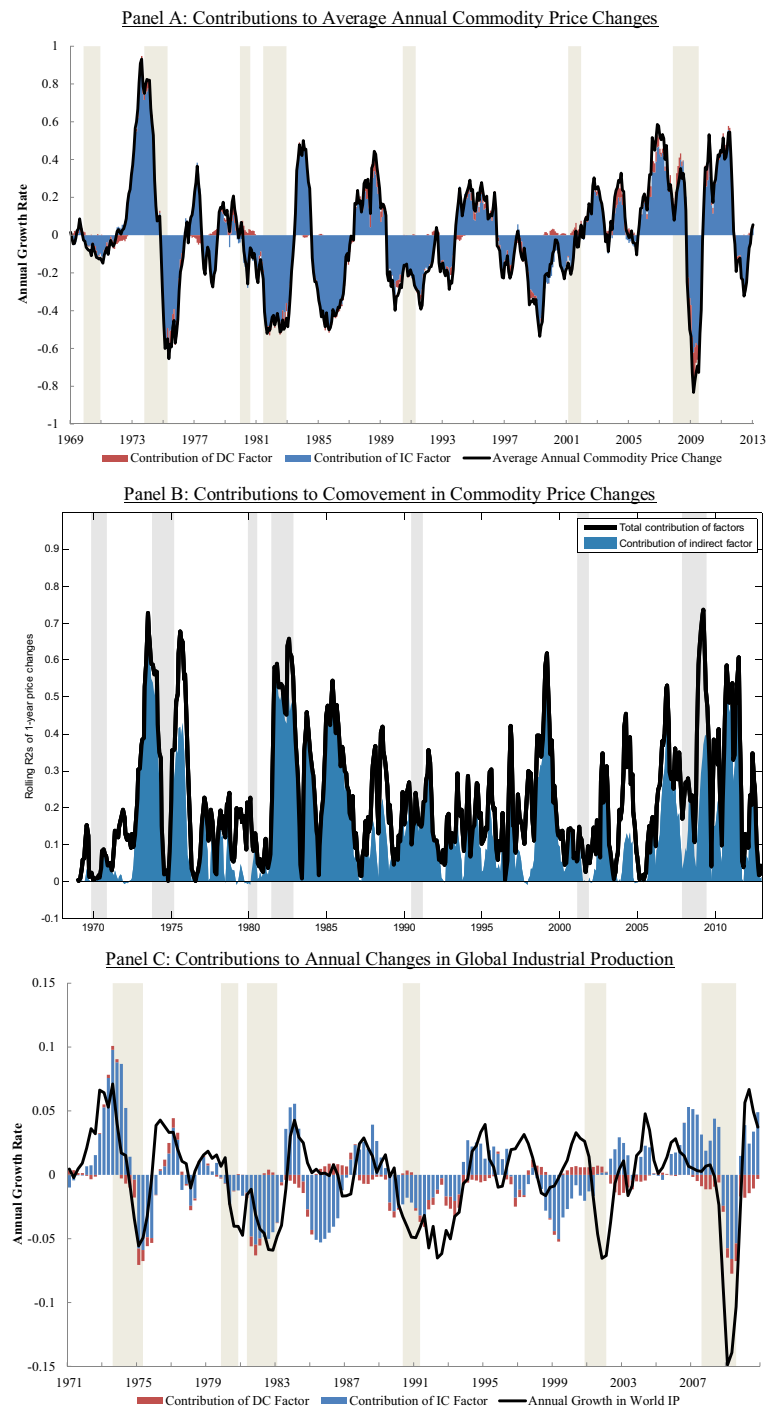


Fig. 2. The contribution of indirect and direct factors to changes in commodity prices.

Notes: Panel A plots the contributions of the direct and indirect factors (DC and IC, respectively) to the average annual price changes across all commodities. Panel B plots the contribution of the two factors to cross-sectional variation in 1-year commodity price changes (black line) and that coming solely from IC factor (blue shaded area). See Section 3.4 for details. Panel C plots the equivalent contributions to the annual growth rate of global industrial production. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

We now assess the contribution of each factor to global economic activity, using the global industrial production (IP) constructed by Baumeister and Peersman (2013) from 1947Q1 until 2010Q4.

Unlike with commodity prices, the factor structure does not immediately lend itself to a decomposition of historical changes in global industrial production. To do so, we first rely on the theory presented in Section 2 in which the IC factor corresponds to the level of global activity that would have occurred in the absence of direct commodity shocks (y_t^{nc}). Because the scale of the IC factor is not identified, we normalize it such that the standard deviation of quarterly changes in the IC is equal to the standard deviation of quarterly percent changes in global IP and treat the resulting historical changes in the IC as the contribution of indirect shocks to global IP. The difference between the demeaned quarterly growth rate of global IP and the demeaned change in the IC (defined as $\delta_t \equiv \Delta y_t - \Delta y_t^{nc}$) should reflect the contribution of direct commodity shocks, potentially omitted factors, and mismeasurement in global production levels. To evaluate the contribution of direct commodity shocks to global IP, we regress δ_t on 4 lags of itself and 8 lags of the direct commodity factor (F_t^{DC}) at a quarterly frequency and construct the contribution of the DC factor to global IP net of the contribution of the IC factor. This approach leaves a component of global activity unaccounted for, potentially reflecting measurement error, omitted variables or model misspecification.

We plot the resulting contributions of the IC and DC factors to global IP growth in Panel C of Fig. 2, again showing only the annual changes to filter out the high-frequency variation in the measurement of global IP. The correlation between the IC factor and global IP is high (0.59) so that historical changes in global IP are primarily attributed to indirect non-commodity shocks. This is particularly true from the early 1970s through the mid-1980s, although commodity-related shocks deepened the decline in global IP during late 1974 and early 1975. As was the case with the decomposition of commodity prices, the decline in economic activity during the Volcker disinflation is accounted for by the IC factor. The dynamics of global activity from the late 1980s to mid-1990s are also largely attributed to the IC factor, although actual changes in global IP exceeded those predicted by the two factors. Growth in the IC factor during the 2000s also coincides with the growth in global IP during this time period, while commodity-related shocks in the DC contributed modest downward pressure on economic activity in 2002 and 2003, then again in 2007–10. To the extent that the DC factor reflects exogenous energy price fluctuations, the negative contribution of the DC factor from late 2007 through 2010 (subtracting 1–2% from the annual growth rate of global IP) is broadly consistent with Hamilton (2009), who argues that oil price shocks contributed to the severity of the Great Recession of 2007–09. Nonetheless, most of the decline in the growth rate of global IP from late 2007 to the depth of the recession can be attributed to declines in the IC factor.

4. Forecasting applications

We examine whether the IC factor contains real-time information relevant for predicting commodity prices, broad commodity price indices and the price of oil in a recursive out-of-sample forecasting exercise. The forecasting model is a monthly linear bivariate FAVAR(p) model for the real price of commodity j and the IC factor.¹⁸ The lag length p is chosen recursively using the BIC. We assess the ability of the IC factor to forecast the 40 individual real commodity prices used to compute the IC factor, three widely used real commodity price indices—the CRB spot index, the World Bank non-energy index and the IMF non-fuel index – and the real price of oil.¹⁹ The forecast performance of the FAVAR is evaluated over two periods. One is commodity-specific and begins either in January 1968 or at the earliest possible date subject to the condition that the initial estimation window contains at least 48 observations. The second one begins in January 1984 and ends in December 2012, with the initial estimation window ending in December 1983. We evaluate the recursive mean-squared prediction error (MSPE) of the FAVAR-based forecast at the 1-, 3-, 6-, and 12-month horizons. All forecast accuracy comparisons are conducted relative to the no-change benchmark. Multiple step-ahead forecasts are computed iteratively using the FAVAR.

Table 4 summarizes the results obtained from the forecasting exercise for the commodity-specific and common sample periods. A summary measure across all commodities is given by the *Aggregate MSPE Ratio* $\equiv \frac{\sum_{j=1}^N MSPE_j^{FAVAR}}{\sum_{j=1}^N MSPE_j^{RW}}$ where $MSPE_j^{FAVAR}$ is the mean-squared prediction error of the FAVAR-based forecast for commodity j ; $MSPE_j^{RW}$ is the mean-squared prediction error of the random walk forecast for commodity j .

For both the commodity-specific and the common forecast evaluation periods, forecasts based on a common factor generate improvements in forecast accuracy between 2% and 8% relative to the no-change forecast up to the 6-month horizon. The FAVAR-based forecasts improve on the no-change forecast for most commodities at short-horizons but performance deteriorates over longer horizons.²⁰ The FAVAR does best at predicting the World Bank non-energy index and the IMF

¹⁸ Because we are unable to reject the null that the rotation matrix equals the identity matrix in Section 3, we use the unrotated first factor in the forecasting exercises.

¹⁹ The IMF non-fuel commodity price index begins in February 1980. The price index was backcast to January 1957 using the IMF agricultural raw material, beverage, food and metals sub-indices with weights obtained from regressing the non-fuel index on the individual sub-indices. The real price of oil is the U.S. refiner's acquisition cost of imported oil. All variables are deflated by U.S. CPI. We apply the EM algorithm recursively to fill in the missing observations.

²⁰ Appendix J provides commodity-specific results. FAVAR-based forecasts generate improvements in forecast accuracy for some agricultural commodities and oils up to 12 months ahead. For the common sample period, 10 (out of 15) of the agricultural commodities and 2 (out of 3) oils achieve improvements

Table 4

Summary of recursive forecast accuracy diagnostics for real commodity prices.

Forecast Evaluation Period: Commodity-Specific										
	Aggregate MSPE Ratio	Distribution of MSPE Ratios					CRB	WB	IMF	Crude Oil
		[0,0.9)	[0.9,0.95)	[0.95,1)	[0,1)	[1,∞)				
1 month	0.921	10	11	11	32	8	0.974	0.834	0.874	0.805
3 months	0.922	4	5	11	20	20	1.057	1.023	0.990	0.977
6 months	0.938	5	4	4	13	27	1.127	1.125	1.072	1.143
12 months	1.096	5	6	5	16	24	1.187	1.214	1.155	1.324
No. of commodities	40						24 (15)	39 (17)	45 (17)	
Forecast Evaluation Period: January 1984–December 2012										
	Aggregate MSPE Ratio	Distribution of MSPE Ratios					CRB	WB	IMF	Crude Oil
		[0,0.9)	[0.9,0.95)	[0.95,1)	[0,1)	[1,∞)				
1 month	0.931	8	7	8	23	5	0.964	0.863	0.888	0.790
3 months	0.944	7	5	6	18	10	0.991	0.982	0.928	0.951
6 months	0.985	8	3	3	14	14	1.068	1.106	1.008	1.114
12 months	1.106	9	2	5	16	12	1.128	1.256	1.112	1.314
No. of commodities	28						24 (15)	39 (17)	45 (17)	

Notes: For the commodity-specific forecast evaluation period, the initial estimation window depends on the commodity. It begins either in January 1968 or at the earliest date that allows the initial estimation window to contain at least 48 observations. The maximum length of the recursive sample is restricted by the end of the data sample for each commodity and the forecast horizon. The “Aggregate MSPE Ratio” is the ratio of the sum of the MSPEs for the bivariate FAVAR forecasts of the real commodity prices relative to the sum of the MSPEs for the no-change forecast. The MSPE ratios of the individual forecasts of real commodity prices are also computed relative to the benchmark no-change forecast. For the FAVAR-based forecasts, the lag length is chosen recursively using the BIC. The number of commodities included in the commodity price indices but not in the cross-section of 40 commodities used to extract the factor is in parentheses.

non-fuel index, with improvements in forecast accuracy up to 17% over short horizons. It also delivers significant improvements relative to the random walk at predicting real oil prices at short horizons.²¹

These findings show that the prices of internationally traded commodities are forecastable in a way suggested by the model presented in Section 2. Improvements in forecast accuracy can be economically important at short horizons, and agricultural commodities and oils tend to be more predictable than industrial commodities. Thus, the factor structure in commodity prices serves a dual purpose for policy-makers and practitioners: it both provides a structural decomposition of the forces driving commodity prices and helps forecast commodity price fluctuations.

5. Conclusion

We propose a new empirical strategy, grounded in a microfounded business cycle model with commodities, to identify the driving forces of global economic activity and commodity prices. The model provides a set of orthogonality conditions and sign restrictions that can be used to identify the parameters of the rotation matrix that yield a structural interpretation of the common factors behind commodity prices, with the “indirect” factor representing the counterfactual level of global economic activity that would have occurred without direct shocks to commodity markets. The IC factor we identify accounts for about 60–70% of the variance in commodity prices, and we cannot reject the theoretical restrictions implied by the model. Its behavior during the 1970s and 1980s suggests that the macroeconomic fluctuations observed during that era were not driven primarily by commodity-related shocks. Nevertheless, there are episodes during which the direct commodity shocks contributed negatively to global economic activity, particularly in the early 1990s and during the Great Recession.

The IC factor can also be used to forecast real commodity prices, some commonly used commodity price indices and the real price of crude oil with a bivariate FAVAR in real-time. Because our identification strategy relies only on commodity prices themselves, it can be implemented for commodities for which market fundamentals are unavailable in real time. In sum, we provide a new conceptual framework for identifying the sources and implications of commodity price comovement and its relationship to global macroeconomic conditions.

in forecast accuracy at the 12-month horizon. The improvements in forecast accuracy in the industrial commodities are concentrated at shorter horizons (i.e., 1- and 3-months).

²¹ Additional results on the ability of the commodity price factor to forecast the real price of oil are in Appendix Table J.3. It compares the bivariate FAVAR with a VAR model of the global oil market that performs well at forecasting the real price of oil out of sample (Baumeister and Kilian 2012; Alquist et al. 2013). The FAVAR model based on the IC factor does well relative to the oil market VAR model at the 1- and 3-month horizons when the BIC is used but it is dominated by the VAR model when a fixed lag length of 12 is used, although the IC factor model still delivers improvements in forecast accuracy up to about 14% relative to the no-change forecast.

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Supplementary materials

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